

Yann Chaubet

Laboratoire Jean Leray, Nantes Université
yann.chaubet@univ-nantes.fr yannchaubet.github.io

RESEARCH INTERESTS

Hyperbolic dynamics, microlocal analysis, spectral geometry, dynamical zeta functions, geometric topology.

POSITIONS

- May 2026 : Visiting researcher, University of Melbourne.
- 2023– : Assistant Professor, Nantes Université.
- 2022–2023 : Herschel Smith Research Fellow, University of Cambridge (By-Fellow of Churchill College).

EDUCATION

- 2019–2022 : PhD in Mathematics, Université Paris-Saclay, supervised by Colin Guillarmou.
- 2014–2019 : École normale supérieure, Paris.

PREPRINTS

- Minimax spectral estimation of weighted Laplace operators (with V. Divol), arXiv:2511.22694.
- Nonlinear chaotic Vlasov equations (with D. Han-Kwan and G. Rivière), arXiv:2407.04426.

PUBLICATIONS

- Dynamical zeta functions, book chapter in *Microlocal analysis in hyperbolic dynamics and geometry*, by Thibault Lefeuvre, published in *Cours spécialisés de la SMF*.
- Combinatorial zeta functions for triangles (with L. Bénard, N. V. Dang and T. Schick), to appear in *Journal of Topology*.
- Geodesic Lévy flights and expected stopping time for random searches (with Y. G. Bonthonneau, T. Lefeuvre and L. Tzou), *Probability Theory and Related Fields* 191 (2025), 235–285.
- Dynamical zeta functions for billiards (with V. Petkov), to appear in *Annales de l'Institut Fourier*.
- Resolvent of vector fields and Lefschetz numbers (with Y. G. Bonthonneau), *Pure and Applied Analysis* 6 (2024), 521–540.
- Dynamical torsion for contact Anosov flows (with N. V. Dang), *Analysis & PDE* 17 (2024), 2619–2681.
- Closed geodesics with prescribed intersection numbers, *Geometry & Topology* 28 (2024), 701–758.
- Closed billiard trajectories with prescribed bounces, *Annales Henri Poincaré* 24 (2022), 1051–1075.
- Poincaré series for surfaces with boundary, *Nonlinearity* 35 (2022).

INVITED TALKS

Mini-courses.

- October 2025, Journée Rennes–Nantes d'analyse, Rennes (2 lectures).
- December 2025, Conférence Géométrie et Dynamique, Aussois (3 lectures).

International conferences.

- Three Days Workshop in Dynamics, Geometry and Analysis, Melbourne, 2026.
- Rigidity Phenomena in Dynamics and Spectral Theory, Zurich, 2024.
- Hyperbolic Dynamics and Semiclassical Analysis, Paris, 2023.
- Analytic Techniques in Dynamics and Geometry, Les Diablerets, 2023.
- Rencontre Aléatoire, Dynamique et Spectre, Grenoble, 2023.
- École de Printemps Géométrie et Dynamique, Lille, 2023.
- Paris–London Analysis Seminar, London, 2023.
- Geometrical Inverse Problems, Linz, 2022.
- Rigidity Problems in Geometry, Roscoff, 2022.
- Spectra and Dynamics on (Locally) Symmetric Spaces, Paderborn, 2022.

Seminars. University of Queensland (2026), Monash University (2026), Pisa (2025), Brest (2025), Tours (2024), Quimper (2024), Nantes (2024), Warwick (2024), Bristol (2023), London (2023), Cambridge (2023), Marseille (2022), IMJ-PRG (2022), Montpellier (2022), Bochum (2022), Pisa (2022), Bordeaux (2022), Orsay (2020–2021), Marseille (2020).

AWARDS AND GRANTS

- 2025 – International Emerging Action (partnership with Melbourne University), 28 k€.
- 2025 – PEPS grant, 4.4 k€.
- 2023 – PULSAR grant, 10 k€.
- 2022 – Herschel Smith Research Fellowship.

CONFERENCE ORGANIZATION

- 2026 – Workshop Analyse semiclassique le long de la Loire, Nantes.
- 2025 – Journées brestoises des jeunes chercheuses et chercheurs en analyse microlocale, Brest.
- 2025 – Three Days Workshop on Geometry, Analysis and Dynamics, Longeville-sur-Mer.
- 2024 – ANR Adyct Workshop, Nantes.
- 2024 – Workshop on Pseudo-Holomorphic Curves, Teyran.